



PLANCK-99

On-device malware detection for embedded Linux.

34 ns latency

37 KB footprint

96.28% accuracy

Zero network dependency

All technical specs sourced from public benchmarks — github.com/Division-36/Planck-99_PublicBenchmarks

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THE PROBLEM

21 billion unprotected devices.

21B

IoT & embedded devices
globally in 2025

[IoT Analytics, Oct 2025]

200K

+

IoT devices hijacked in a
single Chinese state botnet
(Flax Typhoon / Raptor Train)

[FBI Director Wray, Sept 2024]

\$0

production-ready on-device
malware classifier exists today
for MCU-class hardware

[Market gap — no commercial product]

Traditional AV requires 50–500 MB RAM and multi-core CPUs. Embedded hardware provides 256 KB–4 MB. The gap is structural — not a technical limitation.

Three sourced forces converging in 2024-2026

Botnets are now nation-state tools — at IoT scale

Flax Typhoon (PRC-state) compromised 200,000+ IoT devices — cameras, storage, routers — to build the Raptor Train botnet. FBI disrupted it in Sept 2024. Volt Typhoon pre-positioned inside US critical infrastructure for up to 5 years without detection.

[FBI Director Wray, Aspen Cyber Summit, Sept 18 2024 · CISA/NSA/FBI Joint Advisory AA24-038A, Feb 2024]

Regulation now mandates on-device security — legally binding

EU Cyber Resilience Act (2024): mandatory security requirements for all connected devices sold in Europe. UK Product Security & Telecoms Infrastructure Act (2022, enforced April 2024). US NIST SP 800-213A. Perimeter-only defence no longer satisfies compliance.

[EUR-Lex 2024/2847 · UK Legislation 2022 c.46 · NIST SP 800-213A (2022, updated 2024)]

ICS vulnerabilities are at an all-time high and accelerating

2,155 CVEs across 508 ICS advisories in 2025 — the first year CISA published 500+ ICS advisories. Average CVSS score crossed 8.0 in 2024 for the first time ever. 929 high-severity CVEs in 2024 alone. 75% of all published advisories rated high or critical.

[Forescout 'ICS Cybersecurity in 2026', Feb 2026 · SOCRadar ICS Recap, Dec 2025]

Every existing solution fails embedded hardware constraints

Metric	Traditional AV	Cloud Detection	Signature Scanner	Planck-99
Peak RAM	50-500 MB †	Server-side	~30 MB	0.03 MB ★
Inference time	10-100 ms †	200-2,000 ms	1-50 ms	34 ns ★
Network req.	Optional	Mandatory	No	Never ★
GPU required	No	Yes	No	Never ★
Air-gap safe	✗	✗	~	✓ ★
MCU-class	✗	✗	✗	✓ ★
Deterministic	No	No	~	✓ ★
Accuracy (IoT)	95-98% †	N/A	Sig. only	96.28% ★

Source: AV-TEST Institute product comparisons 2024 | † Source: AV-TEST Institute product comparisons 2024 | ‡ Source: AV-TEST Institute product comparisons 2024

Accuracy (IoT) from Division-36/Planck-99 Public Benchmarks | † Source: AV-TEST Institute product comparisons 2024 | ‡ Source: AV-TEST Institute product comparisons 2024

Independent Test 2024

THE SOLUTION

Planck-99

Syscall behaviour analysis · No signatures · No cloud · No GPU · No FPU · No runtime dependencies

DETERMINISTIC

34 ns

$\sigma \leq 14$ ns · 9 runs
100,000 inferences



ULTRALIGHT

37 KB

27 KB binary
+10 KB ring buffer



ACCURATE

96.28%

unseen IoT malware
2016-2026 range



THROUGHPUT

29.4M/s

inferences per second
single MCU core



★ **All metrics:** github.com/Division-36/Planck-99_PublicBenchmarks — public, reproducible, verifiable. Hardware: Intel Xeon 3.8 GHz, 3.8 GB RAM, single CPU core, no GPU, no NPU.

Closed-form syscall analysis — zero black boxes, 100% deterministic

1

Kernel hook

Each syscall appended to 1,200-entry ring buffer. Constant-time op. Window: 500 syscalls minimum for classification.

2

32 features

Shannon entropy · frequency ratios · bigram transitions · temporal burstiness. Every feature has a closed-form mathematical derivation.

3

Normalise

Max-abs normalisation → INT16 quantisation (×1000 scale). No floating-point unit required. Accuracy cost vs float32: ≤0.5%.

4

Random forest

Majority vote across ensemble. Pure INT16 threshold comparisons. Misclassification probability decays exponentially with ensemble size.

5

Output

BENIGN or MALWARE in 34 ns. Slide window 600 syscalls (50% overlap) for continuous real-time monitoring.

Identical input always yields identical output. Each classification produces a JSON proof file — a complete mathematical audit trail of the decision path.

PERFORMANCE — THREE DATASETS, TWO FULLY UNSEEN

All benchmarks public · github.com/Division-36/Planck-99_PublicBenchmarks

BASELINE	UNSEEN ✓	UNSEEN ✓
Training set	LinuxStatic	IoTsyscalls
Accuracy	Accuracy	Accuracy
Precision	Precision	Precision
Recall	Recall	Recall
FPR	FPR	FPR
Internal validation · balanced dataset	Zero false positives · unseen Linux ELF	Primary target · 2016–2026 malware range

Latency vs Python baseline:	Training: Python 17.01 ms → C 62 ns (274,000× faster)	LinuxStatic: Python 17.86 ms → C 35 ns (510,000× faster)	IoTsyscalls: Python 537.65 ms → C 34 ns (15,800,000× faster)
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* FPR note: 12.18% on IoTsyscalls reflects dataset imbalance (1:5.3 benign-to-malware ratio, median trace 2,234 syscalls). On balanced data: FPR = 1.45%. On LinuxStatic: 0.00%. Hardware: Intel Xeon 3.8 GHz, 3.8 GB RAM, single CPU core, no GPU.

\$28B → \$80B market · 18.7% CAGR · No on-device solution exists

TAM

\$28B (2024) → \$80B (2031) · 18.7% CAGR — Global IoT Security Market

Source: MarketsandMarkets, March 2026

Industrial segment: \$26.24B in 2024 · North America: 34.2% of global market share · APAC: 22.2% CAGR (fastest region) · Source: Technavio, March 2026

Why embedded is the uncaptured segment

The industrial OT segment requires air-gapped, on-device detection by definition — no cloud solution qualifies. This is the structural gap Planck-99 fills. SAM and SOM figures are not quoted because no independent sizing study exists for a market with no current product.

[Technavio March 2026 · internal gap analysis]

Regulatory demand is legally binding

EU Cyber Resilience Act: Regulation (EU) 2024/2847 · UK PSTI Act 2022 c.46 (enforced April 2024) · FDA Cybersecurity in Medical Devices Guidance, 2023 · NIST SP 800-213A. Each law mandates on-device security capabilities for connected devices.

[EUR-Lex · legislation.gov.uk · FDA.gov · NIST.gov]

APAC: highest-growth region

APAC IoT security CAGR: 22.2% — highest of any region globally. Japan JC-STAR IoT security labeling (2024) signals regulatory alignment. Rapid industrial digitalization and smart city buildout across Southeast Asia and India.

[Technavio March 2026 · Japan MIC JC-STAR, 2024]

B2B SDK licensing into regulated verticals — three phases

Note: No revenue projections are stated. Pricing to be determined with first design partners. Phases represent directional priorities only.

Phase 1 · Now → 12 months

Design Partners

- 3-5 paid pilot integrations with IoT security gateway vendors
- SDK license model — per-device pricing TBD with first partners
- Target verticals: Industrial OT, edge computing, smart infrastructure
- Success metric: first signed integration agreement

Phase 2 · 12-24 months

Platform Deals

- OEM integration with SoC vendors (NXP, STM, Espressif)
- Chipset-level embed — zero integration effort for end customer
- Medical: FDA Cybersecurity in Medical Devices Guidance (2023)
- Automotive: ISO/SAE 21434 compliance — enforced since 2022

Phase 3 · 24-36 months

Compliance Layer

- EU CRA compliance offering — Regulation (EU) 2024/2847
- CISA reference implementation for ICS/SCADA sectors
- APAC: Japan, South Korea, India — 22.2% CAGR region
- Patent licensing on feature extraction method (filing in Phase 1)

COMPETITIVE LANDSCAPE

We're not improving a product — we're creating a category.

	On-device	Air-gap	Real-time	MCU-class	Notes
Traditional AV (Kaspersky, CrowdStrike)	✗	✗	✓	✗	50–500 MB RAM †
Cloud detection (Darktrace, Vectra)	✗	✗	~	✗	Mandatory connectivity
Signature scanners (YARA, ClamAV)	~	✓	✗	✗	No behavioral analysis
OT platforms (Nozomi, Claroty)	✗	~	~	✗	Network layer only
Planck-99	✓	✓	✓	✓	27 KB · 34 ns · 96.28% ★

★ Planck-99: github.com/Division-36/Planck-99_PublicBenchmarks | † AV-TEST Independent Test 2024 | Competitor features: publicly documented product specifications

One design partner integration. That is the only milestone that matters now.

1

Identify one IoT gateway or OT security vendor

A vendor who ships embedded Linux devices and has an active security team. The product already runs on their hardware class. This is a technical integration, not a product build.

2

Run a live benchmark on their device

Planck-99 deploys as a 27 KB binary with a kernel hook. Integration time is hours, not months. The vendor gets a JSON proof file per classification — auditable and explainable.

3

Sign a paid pilot agreement

A signed agreement is the proof of market demand that no pitch deck can substitute. It de-risks every subsequent conversation with investors, accelerators, and OEM partners.